MAYA MAES-JOHNSON

Baton Rouge , Louisiana, USA
mmaesj1@lsu.edu

EDUCATION

Louisiana State University (LSU) August 2025 - Expected May 2030
PhD Student in Oceanography & Coastal Sciences *Overall GPA: 4.0 / 4.0*
Dissertation: River delta network structure & dynamics

Colorado School of Mines (CSM) December 2024
BS in Applied Mathematics *Overall GPA: 3.4 / 4.0*
Minor in Public Affairs

TECHNICAL SKILLS

Computer Languages	MATLAB, Python, R, UNIX
Modeling Software & Tools	ArcGIS, WRF, TopoTide
Field & Laboratory Skills	ADCP hydrographic surveys, stratigraphic measurement, sediment filtration

PROFESSIONAL EXPERIENCE

Coastal Hydrology, Hydrodynamics, and Oceanography Lab, LSU August 2025 - Present
Graduate Research Assistant; Advised by Dr.Matthew Hiatt

- Studying the evolution of channel network structure and water transport time using hydrodynamic modeling and particle tracking
- Developing methodology to quantitatively analyze the structure and flux behavior of river deltas

Coastal Ecosystem Design Studio (CEDS) Internship, LSU Summer 2026
Summer Intern; Advised by Dr. Traci Birch

- Worked with a team to research and develop restoration design concepts of river siphons in Barataria Bay and strategic vegetation plantings in Neptune Pass
- Conducted field interviews with residents of Plaquemines Parish to integrate community input to design concepts
- Gained experience using Adobe Creative Suite to create visuals for final poster, paper, and presentation

Sedimentology Research Lab, CSM Jun 2023 - December 2024
Undergrad Research Assistant; Advised by Dr. Piret Plink-Björkland, Molly O'Halloran (PhD student)

- Participated in five weeks of fieldwork in Utah to collect stratigraphic data from the early Paleogene rock record
- Extracted profiles of modern rivers exhibiting Froude supercritical flow to study relationship between inertia-driven flows and slope

Jackson School of Geosciences, UT Austin Summer 2024
Research Trainee; Advised by Dr. John Goff, Carson Miller (PhD Student)

- Analyzed and visualized eight years of bathymetric data to observe changes in subaqueous estuarine dunes in Port Aransas, Texas
- Participated in weekly research trainee cohort workshops for professional development

MoWaTer Data Science Fellowship, Baylor University Summer 2022
Data Science Fellow; Advised by Dr. Amanda Hering (Baylor), Dr. Doug Nychka (CSM)

- Collaborated on a team of three students to analyze time-series data and create future predictions of carbon-to-sulfate mass ratio at wastewater treatment plants in Colorado
- Interpreted statistical results and created visualizations to present to stakeholders

HONORS, AWARDS, AND SCHOLARSHIPS

- CSM Undergraduate Research Travel Grant (2024)
- The Hydrographic Society of America (THSOA) Houston Chapter Scholarship (2024)
- Fall 2024 Undergraduate Research Symposium 3rd Place (2024)
- Mines Undergraduate Research Fellow (2024)
- Fall 2023 Undergraduate Research Symposium People's Choice Award (2023)
- Summer Undergraduate Research Fellow (2023)
- McBride Outstanding Sophomore Award (2022)
- McBride Honors Program (2021-2024)
- Thorson First-Year Honors Program (2020-2021)
- Hurliman Foundation Scholarship (2020)

RELEVANT COURSES

Deltaic Geology
Geological Oceanography
Gravity Waves in Shallow Water
Chemical Oceanography
Physics of Sediment Transport
Wetland Hydrology
Physical Oceanography
Modeling the Marine Atmosphere
Scientific Computing

PEER-REVIEWED PUBLICATIONS

1. Strong, S., **Maes-Johnson, M.** (2024). "Nonlinear Dynamics of Mountain Pine Beetle Populations: Discussion of Forestry Policy, a Survey of Existing Mathematical Models, and Code Base Demonstration," CODEE Journal: Vol. 17, Article 8.

CONFERENCE PRESENTATIONS

1. **M. Maes-Johnson**, M. O'Halloran, P. Plink-Björklund (2024), American Geophysical Union (AGU) Annual Conference. "Froude Supercritical Flow Occurrences in Modern Mobile Bed River Systems" (Poster)
2. **M. Maes-Johnson**, C. Miller, M. Davis, D. Duncan, J. Cisneros, J. Goff, S. Gulick, C. Lowery (2024), American Geophysical Union (AGU) Annual Conference. "Impacts of Dredging and Storm Events on Subaqueous Estuarine Dunes in Port Aransas, Texas" (ePoster)

OUTREACH

Environmentors (LSU) August 2025 - April 2026
Mentored a high school student on a project focused on PFAS contamination in water bodies across Baton Rouge

Girls Lead the Way (CSM) January 2024
Instructed a mini-lesson on field measurement in stratigraphy and assisted with flume demonstration during SWE's Girls Lead the Way conference

Nine Mile Canyon Stewardship Day September 2024
Presented on geological history and current research in Nine Mile Canyon, Utah to over 100 Stewardship Day attendees

TRAINING AND WORKSHOPS

CSDMS Earth Surfaces Processes Institute (ESPIn), University of Colorado August 2026